

Bayesian Network Modeling of NFL Game Stats and Market Signals for Value Identification

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May 3, 2026

Problem Statement and Motivation

The problem we are trying to solve is: Does partially overlapping information sources encode independent predictive information? For this problem, the domain and data we've chosen is game stats and market signals for NFL games. Our motivation is to uncover whether line movement (market signals) contains information beyond game stats that reveal whether market pricing is efficient or not. Full efficiency would mean that there are no market signals or game stats that encode information not encoded by both together.

If additional information exists in game stats that is not encoded within market signals, then there is information that could be exploited to further move the line towards efficiency.

Our dataset spans 3,990 games over 15 seasons with 6 market data points per game including opening and closing spreads. The two data sources are joined by season and matchup (home and away teams). Our game stats data is from nflverse[1] and our market data is from SportsBookReviewsOnline.com[2].

Table 1: Sample rows from nfl_game_stats_raw (nflverse)

Season	Wk	Home	Away	H Score	A Score	H EPA/play	H TOs	A TOs
2007	1	IND	NO	41	10	0.055	1	3
2007	1	BUF	DEN	14	15	-0.074	1	1
2007	1	CLE	PIT	7	34	-0.331	5	1
2007	1	GB	PHI	16	13	-0.082	4	1
2007	1	HOU	KC	20	3	0.011	3	3

Table 2: Sample rows from sbr_nfl_odds_raw (SBR)

Season	Home	Visitor	Open Sprd	Close Sprd	Open Tot	Close Tot	ML Home	ML Away
2007-08	Indianapolis	NewOrleans	5.5	5.5	50.5	52.5	-240	200
2007-08	HoustonTexans	KansasCity	38.5	3.0	0.0	37.5	-200	170
2007-08	Buffalo	Denver	3.0	3.0	37.0	38.5	150	-170
2007-08	Cleveland	Pittsburgh	4.5	5.0	37.5	36.0	195	-235
2007-08	Jacksonville	Tennessee	6.5	7.5	37.5	39.5	-350	290

Approach

We plan to implement a BN Bayesian Network to test our hypothesis. The DAG structure Figure 1 is a representation of our assumptions from the domain - historical stats drive the opening line, and the market reacts. Our BN will formally test conditional independence, specifically the causal relationship between line movement and game outcome.

Our baseline will be NB Naive Bayes with assumption that all features are independent. We will use Hill Climbing as the main algorithm to encode our BN model, and we will use Peter-Clark (PC) as a validation check of the dependencies of our current DAG structure. We will test stats and market data in isolation and in combination. Comparing

these three tests will tell us whether stats or market data is adding any signal or if they are redundant. If our BN can't beat Naive Bayes, then it means that the conditional dependencies we're modeling don't provide predictive value, which will formally confirm that closing line is efficient.

Our dataset is dense with signal. First, the EPA per play is the most dense stat in football because it accounts for down, distance, and field position. Other stats, such as yards, turnovers, and scoring efficiency are highly correlated with EPA per play. We will also be able to derive a rolling average of Against-The-Spread (ATS) to substitute the actual ATS - because including the actual ATS would result in data leakage. Derivation of a rolling average ATS will be computed from previous games up to but not including the current game.

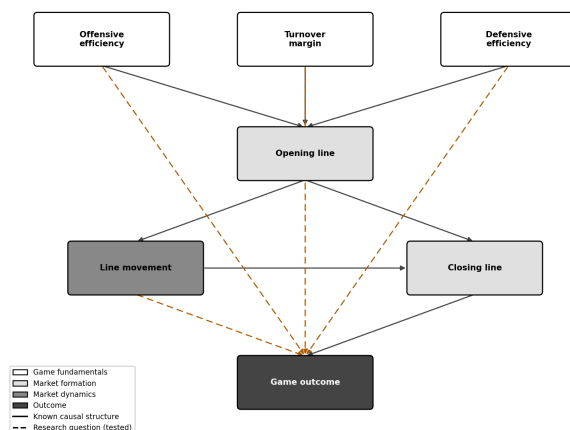


Figure 1: Bayesian network DAG structure. Dashed arrows represent hypothesized direct effects on game outcome — most critically, whether line movement predicts outcome independent of game fundamentals.

Related Work and Novelty

Evaluation Approach

Milestones

[1] <https://nflverse.nflverse.com> [2] <https://sportsbookreviewsonline.com/scoresoddsarchives/nfl/nfloddsarchives.htm>

References